SRIKANTH VELPURI

American Airlines | Amazon

Senior ML/AI Engineer | **End-to-End ML & GenAI Systems** | **ML/AI Platforms** | **MLOps** | **LLMOps** | **Agentic AI**

MS in Data Science – Indiana University Bloomington | BTech – NIT Nagpur

Portfolio | **LinkedIn** | **GitHub** | 812-558-8943 | velpurisrikanth1364@gmail.com | Dallas–Fort Worth, Texas

PROFESSIONAL SUMMARY

Senior AI/ML engineer with 7 years of experience owning ML, GenAI, and agentic AI systems end to end at **American Airlines** and **Amazon**—from framing business problems and building data foundations to training and fine-tuning models, evaluation, deployment, and monitoring—powering ~$100M in annual pricing value and recommendation systems serving ~50K peak QPS. Sets technical standards for security, governance, reliability, and cost, while and driving cross-functional delivery.

EDUCATION

MS in Data Science | Indiana University | Bloomington, IN **01/2021 – 07/2022**

Bachelor of Technology | Visvesvaraya National Institute of Technology | Nagpur, India **07/2013 – 05/2017**

SELECTED IMPACT

- **As a Founding ML Engineer and Forward Deployed Engineer at American Airlines, I** defined and modernized the technical strategy of ML, Data and AI solutions across 350+ airline routes, establishing evaluation-driven release gates and enabling approximately **$100M in annual optimization value**. Recognized for this innovation and business impact through conversion from **contractor to full-time engineer.**
- **Designed ML, data, and AI platform architectures**, service-level objectives, evaluation thresholds, and production-readiness requirements adopted by **100+ AI and ML applications at American Airlines**, increasing release velocity by approximately 25% while maintaining security, reliability, cost, compliance, and governance requirements.
- **At Amazon, led the evolution of an internal recommendation platform** into customer-facing services for Amazon.com US. Designed retrieval, ranking, feature, and inference systems over multi-petabyte data, serving **~50K peak QPS at sub-100 ms p99 retrieval and ranking latency** and 99.99% availability during Prime Day and holiday traffic.

PROFESSIONAL EXPERIENCE

American Airlines | Dallas–Fort Worth, TX **09/2024 – Present**

Senior AI/ML Engineer | Product Engineer - AI/ML

Own ML, GenAI, and agentic AI systems end to end—and the shared ML, data, and AI platforms they run on—supporting solutions tied to ~$100M annual optimization value. Converted to full-time in Dec 2025 based on performance, ownership, and innovation across multiple organizations and teams.

- **Built and deployed governed agentic AI workflows** for yield, route, competitive-fare, and what-if pricing analysis across large-scale demand, inventory, booking-curve, and competitor-fare datasets using custom MCP servers, multi-step orchestration, tiered autonomy, execution budgets, sandboxed tools, fallback policies, and human approval gates, reducing analyst turnaround from four hours to under 15 minutes.
- **Owned forecasting and price-elasticity models end to end**—framed the problem with revenue-management analysts, engineered features from demand and booking-curve data, trained XGBoost, deep learning, and LLMs adapted on analyst-approved pricing decisions, and deployed with monitoring and automated retraining—improving forecast accuracy 12%.
- **Calibrated ML models, AI models, and agents across 350+ airline market routes** through 1,000+ pricing-platform experiments, tuning forecasts, recommendation confidence, and agent autonomy tiers against analyst-approved pricing decisions—lifting recommendation acceptance from 60% to 82%.
- **Built and operated the end-to-end MLOps lifecycle** for forecasting and pricing models—data ingestion, feature engineering, experiment tracking, training, validation, registration, approval, CI/CD, batch and real-time deployment, monitoring, feedback collection, and automated retraining across dev, staging, and production.
- **Shipped an agentic AI pricing platform to production**, integrating LLM-powered agents, tool orchestration, and enterprise APIs to automate market analysis, pricing simulations, and explainable recommendations, supporting 1,000+ pricing experiments across 350+ airline market routes with human-in-the-loop approvals and business guardrails.
- **Deployed and operated multi-cloud model serving and inference** across AWS, Azure, and GCP for traditional ML, deep learning, and LLMs, serving 100+ enterprise applications and 10M+ daily inference requests at 99.9% availability, with ~$30M in potential enterprise impact.
- **Optimized enterprise AI/ML inference performance**, achieving 3× higher LLM throughput, sub-500 ms p95 time-to-first-token (TTFT), 2× higher ML inference throughput, and sub-100 ms p95 latency.

- **Built governed data foundations for ML and GenAI** using Spark, Databricks, Kafka, and Delta Lake, with data contracts, data-quality gates, lineage, and freshness SLAs feeding forecasting, pricing, and agentic workflows.
- **Architected and scaled an agentic AI platform**, establishing a centralized, scalable, and reusable AI infrastructure to standardize AI adoption across business units, empower engineering teams to build and deploy intelligent agents, and accelerate enterprise-wide automation across critical business functions.
- **Established a centralized AI evaluation, observability, decision lineage, and governance framework** incorporating automated evaluation pipelines, LLM-as-judge, distributed agent tracing, production monitoring, and human-in-the-loop controls to standardize AI quality, security, and reliability across enterprise applications and improving operational visibility.
- **Embedded AI security and guardrails** into agentic and LLM applications, including prompt-injection defense, PII protection, sandboxed tool execution, least-privilege agent identity and authorization (OAuth 2.0/OIDC), and AI red teaming aligned with OWASP LLM Top 10 and NIST AI RMF.
- **Led architecture and production-readiness reviews** across data science, ML engineering, platform, security, and revenue-management teams, balancing data volume, model quality, latency, reliability, explainability, governance, and cost while introducing right-sizing, automated shutdown, schedule-to-zero, and team-level cost attribution.

Amazon | Seattle, WA **08/2022 – 09/2024**

Software Engineer—GenAI, Agents & ML/AI Platforms

- **Built and scaled a real-time agentic platform** using Kubernetes, LLMs, RAG, and vector search, supporting 100M+ products, 50K+ requests/sec, and sub-150 ms p99 end-to-end latency.
- **Owned personalization and ranking models end to end**—built two-tower retrieval and DLRM/deep cross-network ranking models, ensured online/offline feature parity, deployed high-QPS gRPC serving, and validated through A/B testing—improving CTR by 9% and conversion by 6%.
- **Owned LLM fine-tuning end to end**—curated training data from user-engagement feedback loops, fine-tuned with LoRA, SFT, and DPO, evaluated with LLM-as-judge and human-label calibration, and deployed with distillation and model routing—cutting serving cost 40% while holding quality within 2% of the base model.
- **Designed multi-agent orchestration** with tool/function calling, contextual memory, semantic and hybrid retrieval, multi-stage ranking, and personalized product discovery; optimized performance through dynamic batching and caching.
- **Calibrated ranking, retrieval, and LLM-judge models** against human labels through online A/B testing and ranking experiments across 100M+ products—using feedback loops from user engagement and failed queries to feed 50K labeled examples/month—improving retrieval quality 15%.
- **Built a distributed GPU inference optimization platform** using Kubernetes (EKS), vLLM, and TensorRT-LLM on AWS across multi-region, multi-tenant GPU clusters, with SLO-aware routing, continuous batching, KV-cache optimization, and workload-aware autoscaling and placement to maximize GPU utilization, reduce latency, and optimize cost.
- **Built an inference optimization framework** integrating model quantization (FP8/INT8), kernel optimization, dynamic benchmarking, and hardware-aware configuration tuning to improve token throughput, reduce GPU memory consumption, and optimize cost per inference while maintaining model quality and p99 latency SLOs.
- **Designed and led an evaluation-driven release framework** for agents, prompts, models, tools, and retrieval systems using LLM-as-judge, human-label calibration, failure replay suites, trajectory scoring, A/B testing, canary promotion, dashboards, and automated rollback gates, reducing production incidents by 35% and shortening release cycles from 2 weeks to 3 days.
- **Developed internal tooling** for experimentation, prompt/config management, feature flags, model comparison, and controlled rollout of GenAI experiences, enabling 10+ teams to ship GenAI changes with 60% fewer production rollbacks.
- **Architected a multi-tenant ML/AI platform** used by 10+ stakeholder groups for dataset contracts, tenant isolation, compute quotas, cost attribution, feature engineering, HPO, validation, deployment, and monitoring.
- **Led design discussions and code reviews for large-scale ML, AI, and agentic systems**, influencing architecture decisions, improving system reliability, strengthening cross-team communication, and mentoring engineers on production best practices.

Flux Auto | Bengaluru, India **01/2018 – 12/2020**

Machine Learning Engineer

- **Owned perception and classification models end to end**—data preprocessing, feature extraction, train/test split strategy, training, evaluation, and experiment comparison—improving detection accuracy from 82% to 94% across baseline iterations.
- **Built automated data-quality checks** for labels, annotation consistency, class imbalance, duplicate frames, and corrupted files, catching 95% of bad samples before training and cutting wasted GPU hours by 30%.
- **Established model evaluation practices**—precision, recall, F1, confusion matrices, false-positive/false-negative analysis, and slice-based testing on edge cases—reducing false negatives in low-light conditions by 25%.
- **Designed retraining workflows** using hard-example mining, targeted data augmentation, and misclassification analysis of failed detection cases, raising edge-case robustness by 20% over 5 iterations.
- **Built repeatable MLOps workflows**—dataset versioning, model-artifact tracking, experiment logging, evaluation reports, and release comparison—cutting experiment reproduction time by 50%.

TECHNICAL SKILLS

AI/ML Architecture & Enterprise Platforms: Enterprise AI/ML architecture, end-to-end AI systems, multi-tenant self-service platforms, control-plane/data-plane design, platform APIs/SDKs, MLOps, LLMOps, model lifecycle management, model registries, feature stores, CI/CD/CT, architecture governance, data/model lineage, responsible AI, scalability and high availability

Agentic AI & Multi-Agent Orchestration: LangGraph, LangChain, OpenAI Agents SDK, Google ADK, MCP, Agent2Agent (A2A), ReAct agents, multi-agent orchestration, supervisor routing, tool/function calling, agentic RAG, GraphRAG, agent memory, context engineering, human-in-the-loop, Temporal, durable execution, state management, prompt engineering and versioning

Generative AI, Foundation Models & Multimodal AI: Azure OpenAI, Amazon Bedrock, Anthropic Claude, Google Gemini, Vertex AI, Azure AI Foundry, Hugging Face Transformers, foundation models, transformer architectures, Mixture of Experts (MoE), vision-language models (VLMs), multimodal RAG, document AI, speech-to-text, real-time voice agents, LoRA/QLoRA, supervised fine-tuning (SFT), DPO, model distillation

RAG, Retrieval & Knowledge Systems: Embeddings, dense/hybrid retrieval, semantic and vector search, metadata filtering, reranking, contextual retrieval, query transformation, grounded generation, knowledge graphs, OpenSearch, Elasticsearch, Qdrant, Pinecone, pgvector, Neo4j, Azure AI Search, retrieval optimization, context-window optimization

ML Platforms, Model Serving & Inference Optimization: MLflow, Kubeflow Pipelines, SageMaker Pipelines, SageMaker Feature Store, SageMaker Model Monitor, Azure ML, NVIDIA Triton, vLLM, SGLang, TensorRT-LLM, KServe, Ray Serve, ONNX, TensorRT, real-time/batch inference, high-QPS serving, autoscaling, model routing, KV-cache optimization, continuous batching, speculative decoding, quantization, inference latency and cost optimization, champion/challenger deployments

AI Evaluation, Observability & Production Reliability: LangSmith, Langfuse, Arize Phoenix, RAGAS, DeepEval, Promptfoo, LLM-as-judge, golden datasets, trace replay, agent trajectory evaluation, task-success evaluation, groundedness and relevance testing, safety evaluation, regression testing, release gates, canary/shadow/A/B testing, drift monitoring, SLOs, error budgets, incident response, automated rollback

AI Security, Governance & Compliance: OWASP LLM Top 10, prompt injection defense, AI red teaming, adversarial testing, AI threat modeling, agent identity and authorization, OAuth 2.0/OIDC, sandboxed tool execution, least-privilege access, AI guardrails, PII protection, responsible AI, NIST AI RMF, ISO/IEC 42001, model governance, auditability and policy enforcement

Cloud, DevOps & Infrastructure Observability: AWS, Azure, GCP, Kubernetes, EKS, AKS, Docker, Helm, Terraform, Argo CD, GitOps, GitHub Actions, Airflow, AWS Lambda, API Gateway, ECS, Glue, Step Functions, Azure Functions, Event Hubs, Key Vault, Entra ID, Grafana, Prometheus, Datadog, OpenTelemetry, CloudWatch, Azure Monitor

Data Engineering, Streaming & Lakehouse Platforms: Spark, PySpark, Databricks, Kafka, Kinesis, Flink, Delta Lake, Apache Iceberg, Snowflake, BigQuery, dbt, lakehouse architecture, batch/streaming pipelines, feature engineering, data contracts, schema evolution, data-quality gates, metadata management, dataset versioning, replayable pipelines, backfills, idempotency, freshness SLAs, data governance and lineage

GPU Infrastructure & Distributed AI Training: NVIDIA GPUs/H100, Ray, DeepSpeed, PyTorch FSDP, Megatron-LM, CUDA, NCCL, Kubernetes GPU scheduling, distributed/multi-node training, tensor/pipeline parallelism, mixed-precision training (FP16/BF16), ZeRO optimization, GPU utilization, workload scheduling, capacity planning, performance profiling, GPU right-sizing

Applied Machine Learning, Recommendations & Forecasting: Recommendation systems, two-tower retrieval, DLRM, deep cross networks, candidate generation, learning-to-rank, personalization, demand/time-series forecasting, price elasticity, dynamic pricing, yield optimization, revenue management, competitive-fare modeling, booking-curve analysis, causal inference, uplift modeling, Bayesian optimization, predictive modeling

Programming, APIs & AI Development Tools: Python, Java, SQL, TypeScript, Go, Bash, PyTorch, TensorFlow, scikit-learn, XGBoost, Hugging Face, FastAPI, REST, gRPC, Pydantic, React, streaming APIs, event-driven microservices, PostgreSQL, DynamoDB, Redis, Playwright, Claude Code, Cursor, GitHub Copilot